

KRAMERS ESCAPE FROM STRAINED VORTICES: THE LEAK LAW, ITS CERTIFIED EXPONENT, AND THE SURVIVAL THRESHOLD

ADILKHAN SALKIMBAYEV

ABSTRACT. A viscous vortex tube in a non-axisymmetric strain whose in-plane strain difference exceeds the axial stretching ($\lambda > 1$) has no steady state. It loses circulation by an exponentially slow leak. Its vorticity is the density of stochastic Lagrangian particles, and the leak is their escape over the saddle of the strain-plus-vortex drift. For a frozen drift we identify the exponent with a Freidlin–Wentzell quasipotential, $\kappa \sim e^{-c(\lambda)/\varepsilon}$ with $\varepsilon = 2\pi\lambda\nu/\Gamma$. We compute $c(\lambda)$ by minimum action ($c(2) = 0.2287$, $c(3) = 0.2098$), certify upper bounds ($c(2) \leq 0.2331$, $c(3) \leq 0.2185$), and derive the large- λ limit $c_\infty = 2\pi \iint_{\text{cat's eye}} (\psi - \psi_s) dA = 0.19258$. The form of the law is due to Moffatt, Kida and Ohkitani (1994) and Prochazka and Pullin (1998), whose constant $1/(2e)$ lies 4.7% below this limit. Direct simulations at $\lambda = 2$ over four decades of κ give a pure exponential, $\kappa \approx 0.53 e^{-0.2153/\varepsilon}$. We prove a one-sided, quantitative version for the true, time-dependent drift (a lower bound on the loss per window, by Girsanov) under a coherence hypothesis. We show that the survival threshold is the leak’s runaway, $\varepsilon_c(T) \approx c/[\ln(CsT) + \ln \ln(CsT)]$, and give numerical evidence for a metastable spectral gap.

1. INTRODUCTION

Stretched vortices are the classical model of the intense filaments of turbulence [11]. A tube of circulation Γ in a uniform strain with axial stretching rate s and in-plane strain difference λs has, for $\lambda < 1$, a steady viscous state: the non-symmetric Burgers vortex (Robinson–Saffman; Moffatt, Kida and Ohkitani [11]; Prochazka and Pullin [12]), whose existence and stability are proved for small λ , and for large circulation, by Gallay and Wayne [8] and Maekawa [10]. For $\lambda > 1$ one in-plane direction is extensional, at rate $s(\lambda - 1)/2$, and no steady state exists. Prochazka and Pullin [12] found that in the regime $\lambda \gg 1$, $\Gamma/\nu \gg 1$ the vortex then decays by an exponentially slow leak through its cat’s-eye boundary,

$$\partial_t \Gamma \propto -\varepsilon^{-1} \Gamma e^{-1/(2\varepsilon)}, \quad \varepsilon = \lambda\nu'/\Gamma',$$

in their units, refining the exponent $e^{-1/(2\varepsilon)}$ of [11]. In ours, $\varepsilon = 2\pi\lambda\nu/\Gamma$ is the ratio of the deviatoric strain $\lambda s/2$ to the peak vorticity $\Gamma s/4\pi\nu$ of the Burgers core.

This paper explains the leak as a large-deviation phenomenon and puts numbers on it. Vorticity in the planar reduction is the density of a diffusion whose drift is the strain plus the vortex’s own velocity (Section 2). The leak is the escape of that diffusion over the two saddles of the drift, a Kramers problem with a non-gradient drift. Our contributions:

- (1) For a frozen drift, the leak exponent is a Freidlin–Wentzell quasipotential barrier, $c(\lambda) = 2\pi\lambda\Delta V'(\lambda)$ (Theorem 3.1).
- (2) The barrier computed by minimum action at finite λ , with certified upper bounds (Section 4).
- (3) The large- λ limit in closed form, $c_\infty = 0.19258$, by streamline averaging (Section 5); Prochazka and Pullin’s $1/(2e) = 0.18394$ is an approximation to it.

- (4) Simulations at $\lambda = 2$ establishing the pure exponential form, $\kappa \approx 0.53 e^{-0.2153/\varepsilon}$, with the measured exponent 6% below the computed one (Section 6).
- (5) A one-sided leak law for the true, time-dependent drift, by Girsanov (Theorem 7.1).
- (6) The survival threshold $\varepsilon_c \approx 0.04\text{--}0.05$ seen in simulations is the leak's runaway on a finite window, not a loss of stability (Theorem 8.1); the shape modes keep a spectral gap up to it (Section 9).

The form $\kappa \propto e^{-c/\varepsilon}$ is not new: it is in [11, 12]. What is new is its identification as Kramers escape, the finite- λ exponent and its certification, the exact large- λ constant, the one-sided theorem, and the runaway threshold. The leak is a building block of a programme on the regularity of the Navier–Stokes equations (the companion papers on self-induction [13], on drift [14] and on circulation in Leray variables [15]).

2. THE TUBE REDUCTION AND THE PARTICLE PICTURE

Consider a straight tube along z in the strain $u_{\text{str}} = (-g_1x, -g_2y, sz)$, $g_1 = \frac{1+\lambda}{2}s$, $g_2 = \frac{1-\lambda}{2}s$. For z -independent vorticity $\omega(x, y, t) e_z$ the Navier–Stokes equations reduce to

$$\partial_t \omega + \nabla \cdot (b\omega) = \nu \Delta \omega, \quad b = -(g_1x, g_2y) + K * \omega, \quad K(x) = \frac{x^\perp}{2\pi|x|^2}. \quad (1)$$

The divergence of b is $-s$ (the axial stretching), so circulation $\Gamma = \int \omega$ is conserved on $\mathbb{R}^2$, and for $\lambda < 1$ the axisymmetric Burgers profile $\frac{\Gamma s}{4\pi\nu} e^{-s|x|^2/4\nu}$ deforms into the non-symmetric Burgers vortex. For $\lambda > 1$ the in-plane rate $-g_2 = \frac{\lambda-1}{2}s$ is extensional. We take $s = 1$ (time in strain units) and write $\varepsilon = 2\pi\lambda\nu/\Gamma$.

The particle picture. Equation (1) is the forward Kolmogorov (Fokker–Planck) equation of

$$dX_t = b(t, X_t) dt + \sqrt{2\nu} dW_t, \quad (2)$$

for any drift b , so $\omega(t)$ is the law of X_t (times Γ) when $X_0 \sim \omega_0/\Gamma$ and b is the drift generated by the solution itself (a McKean–Vlasov diffusion; in three dimensions this is the stochastic Lagrangian picture of Constantin and Iyer [2]). Circulation inside a region is the probability that a particle is there.

Rescaling. With $y = x/\sqrt{\Gamma}$ a point vortex of circulation Γ becomes a unit point vortex, the strain is unchanged and the noise becomes $\nu' = \nu/\Gamma = \varepsilon/(2\pi\lambda)$. In rescaled variables

$$b_\lambda(y) = -(g_1y_1, g_2y_2) + \frac{y^\perp}{2\pi|y|^2}, \quad (3)$$

the model drift. It has a stable spiral at 0 (for a regularised core) and two saddles at $|y|^2 = 1/(\pi\sqrt{\lambda^2 - 1})$ on the line through the origin in the direction where strain and swirl cancel. The basin D of the core is bounded by the stable manifolds of the saddles, and small noise $\nu' \rightarrow 0$ is strong circulation $\Gamma/\nu \rightarrow \infty$.

3. THE EXPONENT AS A QUASIPOTENTIAL

For a drift b and noise $\sqrt{2\nu'}$ the Freidlin–Wentzell action of a path φ on $[0, T]$ is $S_T[\varphi] = \frac{1}{4} \int_0^T |\dot{\varphi} - b(\varphi)|^2 dt$, and the quasipotential of the attractor is $V(y) = \inf\{S_T[\varphi] : \varphi(0) = 0, \varphi(T) = y, T > 0\}$. Minimising over parametrisations gives the geometric form

$$V(y) = \inf_{\varphi} \frac{1}{2} \int_0^1 (|b(\varphi)| |\varphi'| - b(\varphi) \cdot \varphi') d\alpha. \quad (4)$$

Theorem 3.1 (frozen-drift leak law). *Fix $\lambda > 1$ and replace the point vortex in (3) by a Lamb–Oseen core of radius $a' > 0$, small enough that 0 is a stable spiral. Assume the phase portrait (P): D is bounded and ∂D consists of the two saddles and their stable manifolds. Let τ be the exit time of $dY = b(Y)dt + \sqrt{2\nu'}dW$ from D and $\Delta V' = V(\text{saddle})$. Then, uniformly for y in compacts of D ,*

$$\lim_{\nu' \rightarrow 0} \nu' \ln \mathbb{E}_y \tau = \Delta V', \quad \tau / \mathbb{E}_y \tau \rightarrow \text{Exp}(1) \text{ in law,}$$

and the principal Dirichlet eigenvalue $\mu_1(\nu')$ of the generator on D satisfies $\lim \nu' \ln \mu_1 = -\Delta V'$. In physical variables the circulation leak rate per strain time obeys $\lim_{\varepsilon \rightarrow 0} \varepsilon \ln \kappa = -c(\lambda; a')$, $c = 2\pi\lambda \Delta V'$.

Proof. This is the exit problem for a small random perturbation of a flow with one attractor in a bounded domain. Here the boundary is characteristic: it consists of trajectories of the flow (stable manifolds of saddles), so the standard non-characteristic theorem of Freidlin and Wentzell [6, Ch. 4] does not apply directly. The extension to boundaries containing saddle points and their stable manifolds, with the same conclusion $\nu' \ln \mathbb{E} \tau \rightarrow \min_{\partial D} V$, is due to Day [4], and the exponential law to Day [3]. On ∂D the minimum of V is attained at the saddles, since every other boundary point lies on a stable manifold and V is non-increasing along trajectories of b . The eigenvalue statement follows from the exit-time asymptotics because the quasi-stationary distribution exits at rate μ_1 and $\mu_1 \mathbb{E}_y \tau \rightarrow 1$. The leak rate of circulation from the core is the exit rate of the quasi-stationary distribution, by the particle picture. $\square$

Definition 3.2. $c(\lambda)$ is the limit $a' \rightarrow 0$ of $c(\lambda; a')$, computed with the singular drift (3) (Section 4).

Remark 3.3 (what the theorem does not cover). (a) The true drift contains the vortex's own field, which changes as circulation leaks; since κ is exponentially small the change is adiabatic, but a two-sided theorem for slowly varying drifts is not written. Section 7 proves the one-sided bound needed in applications without any adiabatic assumption. (b) As $\varepsilon \rightarrow 0$ the physical core shrinks in rescaled units ($a' \propto \sqrt{\varepsilon}$); the barrier is attained away from the core, so uniformity in a' is expected. Inside $|y| < \delta$ the isotropic inflow is exactly $-y/2$ and the deviatoric correction to V is $O(\delta^2 \varepsilon / \Omega)$, which suggests $|c(\lambda; a') - c(\lambda)| \leq C\delta^2 + o(1)$; Table 1 supports it numerically.

4. COMPUTING THE EXPONENT

Minimum action. We minimise the discrete geometric action (4) from a small circle $|y| = r_0$ around the attractor to the saddle, over polygonal paths with equal-arclength redistribution (L-BFGS), seeded by the rotation-averaged spiral $\dot{\theta} = 1/(2\pi r^2)$, $\dot{r} = r/2$. Its quasipotential $r^2/4$ supplies the start value on $|y| = r_0$ with free angle. Over $r_0 \in [0.03, 0.12]$ and $N \in \{800, 2000\}$ points,

$$c(2) = 0.2287 \pm 0.0003, \quad c(3) = 0.2098 \pm 0.0003.$$

Shooting the characteristics of the Hamilton–Jacobi equation $b \cdot \nabla V + |\nabla V|^2 = 0$ instead gives the same values within 1.4%, with larger scatter. The naive gradient estimate (the barrier of $-\nabla(r^2/4)$ alone) is 0.577; the rotation lowers the barrier by more than half.

The physical cores of Section 6 ($a' \approx 0.06$ – 0.10) move c by less than 1.5%, and upwards.

Proposition 4.1 (certified upper bounds). $c(2) \leq 0.2331$ and $c(3) \leq 0.2185$.

Proof. The action of any explicit path from the attractor to the saddle bounds $V(\text{saddle})$ from above. Take: (i) an inner spiral $r = r_0 e^{\mu(t-t_0)}$, $\dot{\theta} = 1/(2\pi r^2)$, on which the vortex rotation cancels in $\dot{\varphi} - b$ exactly, so the action is $r_0^2(A + B/2)/(8\mu)$ with $A = (\mu + g_2)^2$, $B = 2\mu(g_1 - g_2) + g_1^2 - g_2^2$,

TABLE 1. Uniformity in the core radius ($\lambda = 2$, minimum action, Lamb–Oseen core of rescaled radius a').

a'	0	0.01	0.02	0.03	0.045	0.063	0.080	0.097	0.13
c	0.2309	0.2300	0.2298	0.2295	0.2297	0.2304	0.2323	0.2327	0.2303

plus an oscillatory remainder bounded by $|B|\pi r_0^4/4$ (integration by parts); (ii) the relaxed polygon from the minimum-action computation, whose action we bound by midpoint sums with the Lipschitz estimate $|\partial_s \frac{1}{2}(|b| - b \cdot u)| \leq \|Db\| \leq \max(g_1, |g_2|) + 1/(2\pi r^2)$ (total slack 2×10^{-5}); (iii) the gap between the last vertex and the saddle, enclosed by Newton’s method ($< 10^{-25}$). Evaluated in double precision with an explicit rounding margin of 10^{-10} . $\square$

Gap 4.2. Proposition 4.1 uses floating point with a margin, not directed rounding. Redoing (ii) in interval arithmetic is mechanical.

Corollary 4.3. *For the frozen drift at $\lambda = 2$, $\liminf_{\varepsilon \rightarrow 0} \varepsilon \ln \kappa \geq -0.2331$: the leak is at least $e^{-0.2331/\varepsilon}$ to exponential order.*

The upper bound on c is the direction applications need: a lower bound on the loss.

5. THE LARGE- λ LIMIT

With $z = \sqrt{\lambda}y$ and $\tau = \lambda t$, the model drift (3) becomes

$$\frac{dz}{d\tau} = \nabla^\perp \psi(z) - \frac{z}{2\lambda}, \quad \psi(z) = -\frac{z_1 z_2}{2} - \frac{\ln |z|}{2\pi},$$

and the noise is unchanged. As $\lambda \rightarrow \infty$ the drift is Hamiltonian plus a weak inflow. The level sets of ψ inside the cat’s eye are closed; the separatrix passes through the saddles $|z_s| = \pi^{-1/2}$ on the diagonal $z_1 = -z_2$, where $\psi_s = (1 + \ln \pi)/4\pi$.

Proposition 5.1 (averaged barrier, formal). *Averaging the Freidlin–Wentzell action over the fast rotation along closed streamlines gives $V(\psi) = \frac{1}{\lambda} \int A(\psi)/\oint_\psi |\nabla \psi| dl d\psi$, where $A(\psi)$ is the area enclosed. Since $\oint_\psi |\nabla \psi| dl$ is the circulation inside the streamline, which equals 1 for every closed streamline (the strain is irrotational),*

$$c_\infty = \lim_{\lambda \rightarrow \infty} c(\lambda) = 2\pi \iint_{\text{cat's eye}} (\psi - \psi_s) dA = 0.19258.$$

The integral is evaluated by quadrature in the streamline coordinates, converged in the angular resolution. Minimum action at finite λ gives

λ	2	3	5	10
$c(\lambda)$, minimum action	0.228–0.229	0.209–0.212	0.198–0.200	0.194–0.196
$c_\infty + 0.145/\lambda^2$	0.2288	0.2087	0.1984	0.1940

consistent with Proposition 5.1 and an $O(\lambda^{-2})$ correction.

Remark 5.2 (comparison with Prochazka–Pullin). Prochazka and Pullin’s constant $1/(2e) = 0.18394$ comes from an approximate cat’s-eye vorticity profile (their (3.22)) evaluated at the stagnation point. The exact averaged limit is 4.7% higher. Their table of stagnation-point vorticities cannot separate the two once prefactors are allowed. Their prefactor $\propto \varepsilon^{-1}$ is a second benchmark for the open prefactor problem (Gap 12.2).

Gap 5.3. Proposition 5.1 is formal: a theorem needs the averaging principle for quasipotentials of weakly dissipative Hamiltonian flows (Freidlin–Wentzell [6, Ch. 8] treats the related averaging of the diffusion itself) together with control near the separatrix, where the rotation period diverges logarithmically.

6. SIMULATIONS AT $\lambda = 2$

Method. Equation (1) with $s = \nu = 1$ on $[-12, 12]^2$ (256^2) and, for the smallest ε , on $[-18, 18]^2$ (384^2 , same spacing): finite volumes with MUSCL (minmod) upwind fluxes, Heun time stepping, free-space Biot–Savart by zero-padded FFT convolution, vorticity leaving the box discarded. Initial data: the round Gaussian of radius 2 (the Burgers radius). Point symmetry $\omega(x) = \omega(-x)$ is imposed every step: the core sits at a hyperbolic stagnation point and round-off displaces it at rate $\frac{\lambda-1}{2}$ (the drift of [14]), which is a frame effect in the tube’s own frame. The leak rate is $\kappa = -\frac{d}{dt} \ln f_c$ on $t \in [10, 40]$, with f_c the circulation in $|x| < 4$. *Gates.* With $\lambda = 0.5$ (a steady state exists) the same instrument gives $|\kappa| = 3.0 \times 10^{-10}$ at $\Gamma = 300$ and 5.1×10^{-10} at $\Gamma = 850$, the numerical floor; every rate below is ≥ 600 times it. Refitting on $[20, 40]$ changes κ by 0–2% for $\varepsilon \leq 0.030$, so the rates are not transients. The large box reproduces the small-box rates at $\varepsilon = 0.030$ and 0.020 to 0.07% and 0.2% and corrects $\varepsilon = 0.015$ by -7% .

TABLE 2. Leak rates per strain time, $\lambda = 2$, and the effective prefactor $\ln \kappa + c(2)/\varepsilon$ with $c(2) = 0.2287$.

ε	0.042	0.037	0.033	0.030	0.025	0.020	0.0175	0.015	0.0125
Γ/ν	299	340	381	419	503	628	718	838	1005
κ	8.4e-3	2.5e-3	9.7e-4	4.5e-4	9.5e-5	1.04e-5	2.26e-6	2.98e-7	1.91e-8
$\ln \kappa + c/\varepsilon$	0.67	0.19	-0.01	-0.08	-0.11	-0.04	0.07	0.22	0.52

The form. Over $\varepsilon \leq 0.030$, where the barrier exceeds $7\nu'$, a pure exponential fits the six rates with rms residual 0.070 in $\ln \kappa$:

$$\kappa \approx 0.53 e^{-0.2153/\varepsilon} \quad (\lambda = 2).$$

The fitted exponent is 6% below the computed $c(2) = 0.2287$. With c fixed at 0.2287 the effective prefactor (last row of Table 2) stays within ± 0.25 over these points while κ spans four decades; a power prefactor $\kappa = C\varepsilon^\beta e^{-0.2287/\varepsilon}$ gives $\beta = -0.67$ (rms 0.097). Over a factor 2.4 in ε the functions $\ln \varepsilon$ and $1/\varepsilon$ are nearly collinear, so the data alone cannot pin c ; the theory pins it, and the residual discrepancy is in the prefactor, which the finite core does not explain (Table 1). At $\varepsilon \geq 0.033$ the barrier is only $5\text{--}6\nu'$ and the prefactor rises, as expected outside the asymptotic regime.

$\lambda = 3$. At $\varepsilon = 0.037, 0.033, 0.030$ the rates are $1.2 \times 10^{-2}, 3.5 \times 10^{-3}, 1.5 \times 10^{-3}$, larger than at $\lambda = 2$ as $c(3) < c(2)$ requires; at $\varepsilon = 0.042$ the tube tears.

Mechanism checks. (i) The circulation flux across circles $|x| = r$ is constant for $4 \leq r \leq 8$ and equals κ ($4.4 \times 10^{-4}, 1.03 \times 10^{-5}, 3.16 \times 10^{-7}$ at $\varepsilon = 0.030, 0.020, 0.015$): the leak is a steady outflow through the extensional channel. Beyond $r \approx 3$ the vorticity is depleted along the compressive axis and enhanced along the extensional one; there is no circular stripped edge. (ii) The closed-streamline operator: on each closed streamline of the computed field the averaged steady balance is $\partial_A \ln \omega = -s/(4\pi\nu F(A))$ with the exact geometric factor $F = \Gamma(\psi)T(\psi)/(4\pi A) \geq 1$, $T = dA/d\psi$ [9]; integrated to a self-consistent separatrix, it reproduces the slope $d \ln \kappa / d(1/\varepsilon)$ within 18% with no fitted constant (predicted -0.178 ,

measured -0.217). The magnitude is off by $5\text{--}18\times$, an error located at the separatrix corners, where the boundary-layer closure of [9] is approximate.

7. A ONE-SIDED LEAK FOR THE TRUE DRIFT

Applications need only a *lower* bound on the loss per window, and for the actual drift generated by the solution, which is time dependent. Girsanov's theorem gives this without any adiabatic assumption.

Theorem 7.1 (one-sided leak). *Let $b(t, y)$ be a drift on $[0, T] \times \mathbb{R}^2$, let $\varphi \in H^1([0, T]; \mathbb{R}^2)$ and $\delta > 0$, and suppose that on the tube $N_\delta = \{(t, y) : |y - \varphi(t)| < \delta\}$*

- (i) *b is L -Lipschitz in y ;*
- (ii) *(H) $|b - b_{\text{model}}| \leq \eta$, with b_{model} any reference drift.*

Let X solve (2) with noise $\sqrt{2\nu}$ and $|X_0 - \varphi(0)| \leq \delta/2$, and put $A = (\sqrt{S_T[\varphi]} + \eta\sqrt{T}/2)^2$, $S_T[\varphi] = \frac{1}{4} \int_0^T |\dot{\varphi} - b_{\text{model}}(t, \varphi)|^2 dt$. Then

$$\mathbb{P}\left(\sup_{t \leq T} |X_t - \varphi(t)| < \delta\right) \geq q_0 \exp\left(-\frac{A}{\nu} - \sqrt{\frac{2A}{\nu q_0}}\right),$$

where $q_0 = c_0 \exp(-j_{0,1}^2 \nu T / \delta^2 - L^2 \delta^2 T / 4\nu - \sqrt{L^2 \delta^2 T / 2\nu} e^{j_{0,1}^2 \nu T / 2\delta^2} / \sqrt{c_0})$, $j_{0,1} = 2.405$ and $c_0 > 0$ is absolute. With $\delta^2 = \nu/L$, q_0 depends on LT only.

Proof. Put $Y = X - \varphi$, $h(t) = b(t, \varphi) - \dot{\varphi}$ and $r(t, Y) = b(t, \varphi + Y) - b(t, \varphi)$, so that $dY = (h + r) dt + \sqrt{2\nu} dW$ and $|r| \leq L|Y|$ on N_δ . Let $E = \{\sup_{t \leq T} |Y_t| < \delta\}$.

Step 1 (Girsanov for h). Under $\mathbb{Q}$ with $\frac{d\mathbb{P}}{d\mathbb{Q}} = \exp\left(\int \theta \cdot d\tilde{W} - \frac{1}{2} \int |\theta|^2\right)$, $\theta = h/\sqrt{2\nu}$, we have $dY = r dt + \sqrt{2\nu} d\tilde{W}$, and $\frac{1}{2} \int |\theta|^2 = S/\nu$ with $S = \frac{1}{4} \int |h|^2$. By Jensen's inequality conditional on E ,

$$\mathbb{P}(E) = \mathbb{Q}(E) \mathbb{E}_{\mathbb{Q}}\left[e^{\int \theta \cdot d\tilde{W} - S/\nu} \middle| E\right] \geq \mathbb{Q}(E) \exp\left(\mathbb{E}_{\mathbb{Q}}\left[\int \theta \cdot d\tilde{W} \middle| E\right] - \frac{S}{\nu}\right),$$

and since θ is deterministic, Cauchy–Schwarz gives $\mathbb{E}_{\mathbb{Q}}[\int \theta \cdot d\tilde{W} | E] \geq -(\int |\theta|^2)^{1/2} \mathbb{Q}(E)^{-1/2} = -\sqrt{2S/\nu} \mathbb{Q}(E)^{-1/2}$.

Step 2 (small ball). A second change of measure removes r , under which Y is $\sqrt{2\nu}$ times a planar Brownian motion; on E , $\frac{1}{2} \int |r|^2 / 2\nu \leq L^2 \delta^2 T / 4\nu$. The same Jensen–Cauchy–Schwarz argument gives $\mathbb{Q}(E) \geq R(E) \exp(-L^2 \delta^2 T / 4\nu - \sqrt{L^2 \delta^2 T / 2\nu} R(E)^{-1/2})$, where $R(E)$ is the probability that Brownian motion with generator $\nu \Delta$, started within $\delta/2$ of the centre, stays in the disc of radius δ up to time T . By the eigenfunction expansion, $R(E) \geq c_0 e^{-j_{0,1}^2 \nu T / \delta^2}$. Hence $\mathbb{Q}(E) \geq q_0$.

Step 3 (the model). $h = (b_{\text{model}}(t, \varphi) - \dot{\varphi}) + (b - b_{\text{model}})(t, \varphi)$, so by Minkowski $\sqrt{S} \leq \sqrt{S_T[\varphi]} + \eta\sqrt{T}/2$, i.e. $S \leq A$. Combining the steps gives the claim. $\square$

Corollary 7.2 (loss per window). *Let $\omega \geq 0$ solve (1) with its own drift b , let $\mathcal{C}$ be a core region, and let φ run from a point of $\mathcal{C}$ to a point at distance $> \delta$ outside $\mathcal{C}$ on the outflow side of a saddle. If (i), (ii) hold along φ and $\int_{|y - \varphi(0)| < \delta/2} \omega_0 \geq m \Gamma_c(0)$, then at least the fraction $m q_0 e^{-A/\nu - \sqrt{2A/\nu q_0}}$ of the core circulation lies outside $\mathcal{C}$ at time T (plus whatever was supplied).*

Proof. Given the solution, b is a fixed function of (t, y) and ω solves the linear Fokker–Planck equation with that drift, so $\omega(T)$ is Γ times the law of X_T . Apply Theorem 7.1 to each starting point. $\square$

Size of the bound. With b_{model} the rescaled point vortex plus strain and φ the certified path of Proposition 4.1, re-timed to a window $T \approx 6/s$, the exponent is $2\pi\lambda S_T/\varepsilon = 0.236/\varepsilon$ at $\lambda = 2$, against the geometric $0.2331/\varepsilon$. The bound is exponentially sharp up to the coherence error η .

Remark 7.3 (what (H) is). Hypothesis (H) is the definition of a *coherent* tube in the tearing regime: its self-generated drift near the core and the saddle stays within η of the point-vortex-plus-strain model. A torn tube violates (H), but it has then already lost its core.

8. THE SURVIVAL THRESHOLD IS THE LEAK'S RUNAWAY

Simulations at $\lambda = 2$ show quasi-steady cores for $\varepsilon \lesssim 0.04$ and destruction within $t = 30$ for $\varepsilon \geq 0.05$: core fraction at $t = 30$ equal to 0.952, 0.845, 0.042, 0.000 at $\varepsilon = 0.03, 0.04, 0.05, 0.06$. At $\varepsilon = 0.05$ the loss is not exponential ($f_c = 0.82, 0.59, 0.04$ at $t = 10, 20, 30$): it accelerates.

Theorem 8.1 (runaway). *Suppose the core obeys $\dot{\Gamma}_c = -\kappa(\varepsilon)s\Gamma_c$ with $\kappa = Ce^{-c/\varepsilon}$ and $\varepsilon = 2\pi\lambda\nu/\Gamma_c$, and that it tears once $G = c/\varepsilon$ falls to $G_* = O(1)$. Then $\text{Ei}(G_0) - \text{Ei}(G(t)) = Cst$; the lifetime is*

$$sT_{\text{life}} = \frac{\text{Ei}(c/\varepsilon_0) - \text{Ei}(G_*)}{C} = \frac{\varepsilon_0}{c\kappa(\varepsilon_0)}(1 + O(\varepsilon_0)),$$

and the survival threshold over a window T , defined by $T_{\text{life}} = T$, solves $\text{Ei}(c/\varepsilon_c) = C sT + \text{Ei}(G_*)$, so that $\varepsilon_c(T) = c/[\ln(CsT) + \ln \ln(CsT) + o(1)]$ as $sT \rightarrow \infty$.

Proof. $G \propto \Gamma_c$, so $\dot{G} = -CsGe^{-G}$, i.e. $\frac{e^G}{G}dG = -Csdt$, and $\frac{d}{dG}\text{Ei}(G) = e^G/G$. The asymptotics follow from $\text{Ei}(G) = \frac{e^G}{G}(1 + G^{-1} + O(G^{-2}))$. $\square$

With the fitted constants ($c = 0.2153$, $C = 0.53$, $G_* = 1$) the threshold is $\varepsilon_c = 0.085, 0.056, 0.040, 0.026, 0.014$ for $sT = 10, 30, 100, 10^3, 10^6$. Checks: the pre-runaway rate at $\varepsilon = 0.05$ ($t = 10$ – 20) is $\kappa = 0.033$, so $\kappa T = 0.99$ at $T = 30$; the formula's $\varepsilon_c(30) = 0.056$ is 15% above the observed threshold between 0.04 and 0.05. The model is crude near the end, because leaked vorticity comes off the edge and the protective core's effective circulation falls more slowly than f_c , but the mechanism is clear.

Remark 8.2. Survival and leak are one law at two horizons. A threshold such as $\Gamma_c/\nu \gtrsim 2\pi\lambda/\varepsilon_c$ is the leak over a finite window, and over an infinite (for instance self-similar) history it becomes the statement that the circulation lost per unit of strain time is a fixed positive fraction $\kappa(\varepsilon)$, which no finite Γ_c removes.

9. METASTABILITY: A SPECTRAL GAP

Linearise (1) about the computed quasi-steady vortex ($\lambda = 2$). The centroid obeys $\dot{x}_c = Dx_c$ exactly and the equation is invariant under shifts along solutions of it [14], so position decouples from shape: the translation modes are exactly $+\frac{\lambda-1}{2} = +0.5$ and $-\frac{\lambda+1}{2} = -1.5$. Leading eigenvalues (ARPACK on $e^{0.5L}$, centred scheme, base state relaxed under the same scheme):

ε	drift mode	mass (leak) mode	first shape mode	x -translation
0.015	+0.50000	0.00000	−0.974	−1.50005
0.030	+0.49948	+0.00256	−0.853	−1.49859
0.040*	+0.4946	+0.024	−0.768	−1.454

(*base state not relaxed; the mass and translation entries carry a base-state artefact.) The shape modes relax at $\geq 0.77s$ up to $\varepsilon = 0.04$, and the gap shrinks slowly as ε grows (0.97, 0.85, 0.77). It is nowhere near closing before ε_c : the threshold is the runaway of Section 8, not a spectral instability.

Conjecture 9.1 (metastable gap). *For $\lambda \in (1, 3]$ there is $\varepsilon_1 > 0$ such that for $\varepsilon < \varepsilon_1$, in the centroid frame and in a Gaussian-weighted space localised inside the separatrix with absorbing boundary, the linearisation has one eigenvalue of size $O(e^{-c(\lambda)/\varepsilon})s$ (the leak) and the rest of its spectrum in $\text{Re } \sigma \leq -\frac{1}{2}s$.*

Heuristic. At $\lambda = 0$ the spectrum is known (Gallay–Wayne [8]): radial modes at $-ks$, translations at $-s/2$, mass at 0, and the non-radial part moves left as circulation grows (Maekawa [10]; enhanced dissipation, Gallay [7]). In the core the deviatoric strain is small relative to the rotation, $e/\omega_0 = \varepsilon$, and the absorbing boundary turns the mass mode into the quasi-stationary eigenvalue of Theorem 3.1. A proof needs the strain controlled relative to the skew rotation part, which standard perturbation theory does not do, and an equilibrium to linearise about, which for $\lambda > 1$ exists only in the localised (quasi-stationary) sense.

10. OTHER KERNELS

Replace K by $K_\sigma(x) = x^\perp |x|^{2\sigma-2}/2\pi$ ($\sigma = 0$ is Biot–Savart; $\sigma > 0$ less singular). Rescaling $y = x/\ell$ with $\ell = \Gamma^{1/(2-2\sigma)}$ again gives a unit vortex in the strain, with noise $\nu\Gamma^{-1/(1-\sigma)}$, so

$$\ln \kappa \approx -\Delta V'(\lambda, \sigma) \Gamma^{1/(1-\sigma)} / \nu.$$

At $\lambda = 2$, $\Delta V' = 0.0305, 0.0184, 0.0084$ for $\sigma = -\frac{1}{4}, 0, \frac{1}{4}$ (shooting; about 10% high). At $\sigma = \frac{1}{4}$ the simulated rates are 2.8×10^{-6} to 1.3×10^{-8} over $\varepsilon = 0.042\text{--}0.030$, $10^3\text{--}10^4$ times below $\sigma = 0$, with local slope $d \ln \kappa / d\Gamma = 0.057$ against the predicted 0.066; at $\sigma = -\frac{1}{4}$ no core survives in that window. Only for $\sigma = 0$ is ε invariant under the parabolic scaling $x \sim \sqrt{\nu t}$, Γ fixed, so only for Navier–Stokes is the leak per unit of strain time a scale-invariant constant along a self-similar compression.

11. REMARKS ON THREE DIMENSIONS

Tilting. For $\lambda > 3$ the tube axis is unstable to tilting (Prochazka and Pullin [12]); after realignment with the new maximal extension the tube has $\lambda' = (\lambda + 3)/(\lambda - 1) \in (1, 3)$ with axial strain $s(\lambda - 1)/2$. Tilting therefore maps every tube into $1 < \lambda \leq 3$, where the leak law applies with λ' . *Short waves.* A short-wave (WKB) scan of the quasi-steady vortices at $\varepsilon = 0.030\text{--}0.015$, with axial stretching, gives a maximal viscous gain $e^{1.9}$ over $t \leq 10/s$; at these circulations viscosity bounds the elliptic instability, in line with the stretching-limited resonance of Eloy and Le Dizès [5]. The leak is the dominant loss mechanism in this range.

12. OPEN PROBLEMS

Gap 12.1 (two-sided law for the true drift). The asymptotics $\varepsilon \ln \kappa \rightarrow -c(\lambda)$ for the self-consistent, slowly varying drift (an adiabatic Freidlin–Wentzell theorem). Theorem 7.1 gives the lower bound on the loss.

Gap 12.2 (prefactor). The non-reversible Eyring–Kramers prefactor (Bouchet and Reygnier [1]) at the rotating attractor and the saddles, to compare with the measured $C\varepsilon^{-0.67}$ (at $c = 0.2287$) and with Prochazka and Pullin’s ε^{-1} . Two attempts by Riccati integration along the minimum-action path failed to converge near the saddle; a backward integration from the saddle or a direct eigenvalue computation is needed. Structural fact: $\text{div } b = -1$ exactly and the Hessian of V at the attractor has trace 1, so the prefactor is ε -independent at fixed core.

Gap 12.3. Interval arithmetic for Proposition 4.1; the averaging theorem behind Proposition 5.1; Conjecture 9.1. The hypothesis (P) of Theorem 3.1 is verified numerically, not proved.

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ALMATY, KAZAKHSTAN

Email address: `adilsalkimbayev@gmail.com`